

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

(0-4 AT-2)

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total(10)
Weightage	20%	25%	25%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I S. Chandra Sheela (1923 72/63) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Chandra Sheela

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

	Customer Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

Q2. Dataset: Credit Card Transactions

	Transaction Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

Q3. Dataset: Telecom Usage

User		Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200	
U2	100	2	100	
U3	350	6	220	
U4	80	1	90	
U5	320	5.5	210	

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

Q5. Dataset: Traffic Data

Location Vehicle Count Avg Speed (km/h) Delay (min)

L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

Dataset: Supermarket purchase

Customer	visit/month	Avg spend	items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

The Given dataset contain three Numerical features

- Customer Visit per month
- Average spend
- Number of item Bought

A suitable approach is k-means clustering. Before clustering the Numerical values should be Normalized because avg spending value much larger than other variable

using $k=2$ the customer can be segmented as follows

Cluster

Customer

Characteristic

Frequent
budget shopper

C1, C3, C5

Visit frequently Spend moderately, buy more items

perman occasional
shopper

C2, C4

visit less frequency, spend more per visit buy fewer item.

Interpretation

Cluster 1 - frequent Budget shoppers:-

Customer C1, C3, and C5 visit the Supermarket frequently and purchase a large number of items, but their average spending is relatively moderate.

cluster-2 - premium occasional shoppers:-

customer c2 and c4 visit less frequently but send a higher amount per visit. They can be targeted with premium product recommendations personalized offers more frequencies visit.

Conclusion

k-mean clustering help the supermarket identify different customer purchasing patterns. By segmenting customer based on visits, spending and no of items purchased, the supermarket can implement target market strategies.

(2)

Dataset: Credit Card Transaction

Transaction	Amount	Location	time
T ₁	200	Chennai	10
T ₂	5000	Delhi	2
T ₃	150	Kolkata	11
T ₄	7000	Mumbai	1
T ₅	180	Chennai	12

To detect unusual or potentially fraudulent transaction we can use k-means clustering or DBSCAN. Since fraudulent transaction are often very different from normal transaction, clustering helps identify transaction do not belong normal behaviour pattern.

Step 1: features used

- The transaction data contains:
 - Transaction Amount
 - Location
 - Transaction time

Since location is categorical data, it should first be converted into Numerical form using Label encoding or one-hot encoding. The Numerical Features should be Normalized

Step 2:- Clustering the transaction

The transaction can be grouped as follow

Cluster	Transaction	pattern
Normal cluster	T ₁ , T ₃ , T ₅	low-value transaction in Chennai
unusual cluster	T ₂ , T ₄	during daytime Very high-value transaction at unusual times

Step-3 identify Suspicious Transaction

- T₁, T₃, T₅ Similar because they have low transaction (150-200) occur in Chennai, and happen b/w 10-12 hours
 - T₂ has a high transaction amount of 5000 and occur at 9am in Delhi
 - T₄ has a very transaction amount of 7000 and occur at 1am in Mumbai
- ∴ T₂ and T₄ Can be Considered unusual or potentially fraudulent transaction because they differ significant from transaction cluster.

Conclusion

A clustering algorithm can automatically group similar transaction and identify transaction that are far from Normal transaction pattern in this dataset, T₂ and T₄ are potential fraud cases due to their high transact

③

Dataset: Telecom Usage

user	call duration	Data usage	Recharge
U1	300	5	200
U2	100	2	100
U3	250	6	220
U4	80	1	90
U5	320	5.5	210

clustering - Based Identification of churn-prone Telecom users

- call duration
- data usage
- Recharge Amount

A suitable algorithm is k-mean clustering

Step-1 preprocess the Data

The Features have different scales, so the data should be Normalized or standardized before applying clustering

Step-2 form Customer clusters

using $k=2$, the users may be group as follow

cluster

users

Characteristics

High-usage
user

U1, U3, U5

High call duration, high data higher
recharge

low usage / churn
prone user

U2, U4

low call duration, low data usage
low recharge.

Step-3 identify Churn-prone users

User U2 and U4 have

- low call duration
- low data usage
- low recharge amount

These users show lower engagement with the telecom service. Therefore they may be considered potential Churn-prone users

Customer Retention strategies

The telecom company can target low-usage cluster using:

- special recharge offer and discounts
- extra data and call benefits
- personalized plans based on usage
- loyalty rewards
- Promotional message to increase engagement

High-usage customer such U1, U3 and U5 can be given premium plans and loyalty benefits to maintain their satisfaction

Conclusion

clustering helps telecom companies divide user into groups based on their behaviour. in this dataset, U2 and U4 can be identified as churn-prone users because of their low usage and recharge patterns

Target offers and personalized retention strategies can help increase engagement and reduce churn customer

④ Data Set :- E-commerce Behaviour

User	product views	clicks	purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Basic observations

- Highest views: U3 with 60 views
- Highest clicks: U3 with 25 clicks
- Highest purchase: U3 with 6 purchases
- Total views: 215
- Total clicks = 85
- Overall click-through rate: $85 / 215 \times 100 = 39.53\%$
- Overall purchase conversion from clicks: $19 / 85 \times 100 = 22.35$
- Overall purchase rate from views: $19 / 25 \times 100 = 8.84$

Thus U3 is the most achieve and active and valuable user in the dataset, having the highest number of views, clicks and purchase

1. Clustering technique: k-means

k-means divides users into predefined number of cluster by minimizing the distance b/w users and their centers

example, we can choose $k=2$.

- cluster 1 - Highly active users: U1, U3, U5

These users have relatively views, click and purchase

- cluster 2 - less active users

These users have lower engagement and lower purchase

② How clustering improves recommendation system

- personalized recommendation:- user in the same cluster have similar shopping behaviour, so product preferred by one user can be recommended to other user in that cluster

- Better product targeting:- Highly active users can receive recommendations for related or premium products while less active users can receive popular or introductory products

- improved accuracy:- Recommendations are based on behavior similarity rather than treating every user identically

Conclusion

K-means clustering can effectively group customer according to their e-commerce behavior. By identifying group of users with similar viewing clicking and purchasing pattern a recommendation potentially increasing user engagement and purchase.

⑤ Dataset: Traffic Data

Location	Vehicle count	Avg Speed	Delay
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Clustering - Based solution For traffic congestion

A suitable technique for this dataset is k-means clustering. It can group location based on their vehicle count, Average speed and Delay.

1. Features used

The clustering algorithm uses:

- Vehicle count - Number of vehicles at the location
- Average speed - lower speed indicates heavier congestion
- Delay - higher delay indicates more congestion.

The features should be normalized before applying k-means because they different numerical range.

② Apply k-means

Choose $k=2$ to create two groups

Location	Vehicle count	Avg Speed	Delay	possible cluster
L1	200	30	10	low congestion
L2	500	15	25	
L3	220	28	12	high congestion
L4	600	10	30	

Low - congestion Zones: L₁, L₃, L₅

High - congestion Zone: L₂, L₄

L₂ and L₄ have high vehicle counts, low average speeds, and high delays making them the major congestion zones.

3. How it helps traffic management

1. identifies congestion hotspots
2. Better traffic signal control:
3. Traffic diversion:
4. Resource allocation:
5. infrastructure planning:

Conclusion

K-means clustering provides an effective way to classify road location into congestion zones using vehicle count speed and delay. This helps traffic authorities make data-driven decision.